Lab 4 session

Harris Corner Detector:

Basic steps for applying the Harris corner detector:

1. Smoothing of the selected gray image using a Gaussian filter.

Sigma = 1, size of [5, 5]. See notes file (table 1)

2. Calculating the first partial derivatives (first order) of the smoothed image with respect to x and y (the horizontal and vertical components of the gradient) using Prewit filters and the convolution product, and a threshold of 100 (to convert gradients into binary images). See notes file (table 2)
3. Calculating the second partial derivatives (of second order) of the previous form with respect to x, y, and x & y altogether using equations (1, 2 and 3 in notes file).
4. Calculating the second moment matrix components (M_{11}, M_{12}, M_{21}, M_{22}) using another Gaussian filter (w) and convolution process.

Sigma = 2, size of [13, 13]. See notes file (table 3)

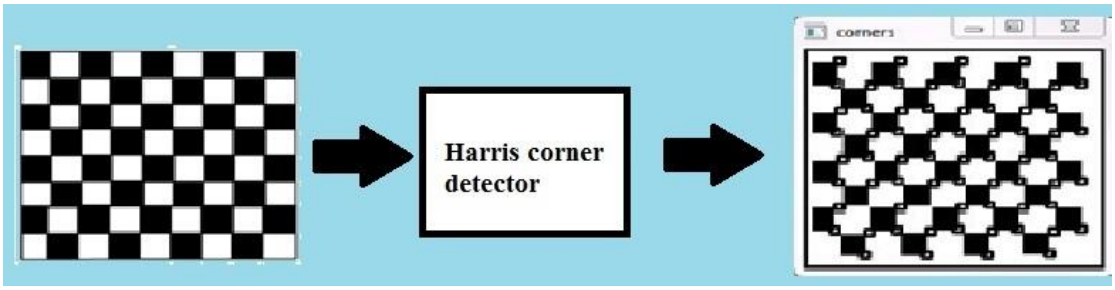
5. Finding interest points using the Harris criterion for important points/ feature:

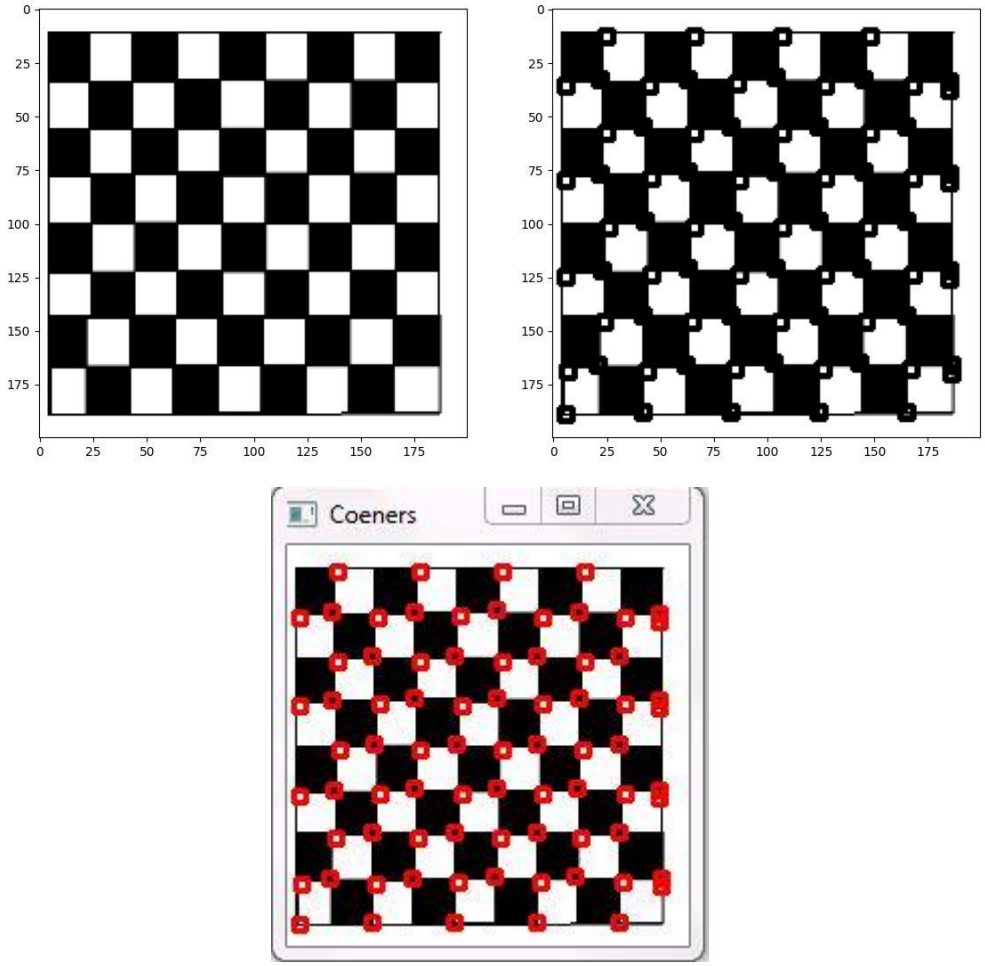
$$R = \frac{\det M}{trace\ M}$$

det (M) = M_{11} . M_{22} – M_{12} . M_{21}
trace (M) = M_{11} + M_{22}
Make sure that all NaN values become zeros.

6. Applying the non-maximum suppression process by keeping the value of current central pixel if it is equal to the max value of the [7, 7] window around it, or eliminate it.
7. Normalize the result of non-maximum suppression to become in the range [0, 1].

8. Apply thresholding process on the normalized result using a threshold of 0.35.
9. Draw a bounding box surrounding the corners.





الشكل (1). نتيجة تطبيق كاشف Harris على صورة لوح شطرنج